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On the Capacity-Achieving Distribution of the Discrete-Time Non-Coherent and Partially-Coherent AWGN Channels

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Abstract

We investigate the ultimate communication limits over rapid phase varying channels and consider the capacity of a discrete-time non-coherent additive white Gaussian noise channel under the average power constraint. We obtain necessary and sufficient conditions for the optimal, capacity achieving, input distribution and show that the optimal distribution is discrete and possesses an infinite number of mass points. Using this characterization of the capacity achieving distribution we compute a tight lower bound on the capacity of the channel based on examining sub optimal input distributions. In addition, we provide some easily computable lower and upper bounds on the channel capacity. Finally, we extend some of these results to the partially coherent channel, where it is assumed that a phase locked loop (PLL) is used to track the carrier phase at the receiver, and that an ideal interleaver and de-interleaver are employed - rendering the Tikhonov distributed residual phase errors statistically independent from one symbol interval to another.

Keywords: non-coherent channel, partially-coherent channel, channel capacity, capacity achieving distribution, discrete, Kuhn-Tucker conditions.

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